

Correct Answers, Uneven Grounding

THE DISTRIBUTED-EVIDENCE GAP IN LEGAL RAG

- Answers stay level.
 - ◆ Grounding falls when evidence spans pages.
- Similar answer quality does not guarantee complete source support.

1 HYPOTHESIS

RESEARCH QUESTION

How does distributed evidence affect the relationship between answer quality and source grounding in a legal RAG pipeline?

H1

Multi-page questions will retain comparable answer scores but show lower final page grounding; uniform citation narrowing may widen the gap.

STUDY COHORT

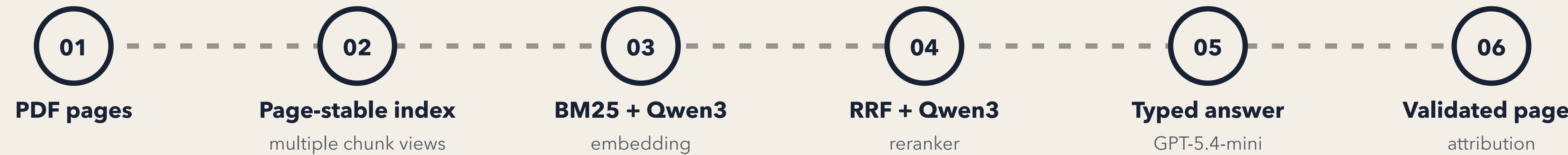


100-question benchmark
95 eligible questions
with annotated source pages
Page F-beta, beta = 2.5 · recall-weighted

Exploratory retrospective slice analysis
of one saved development run.

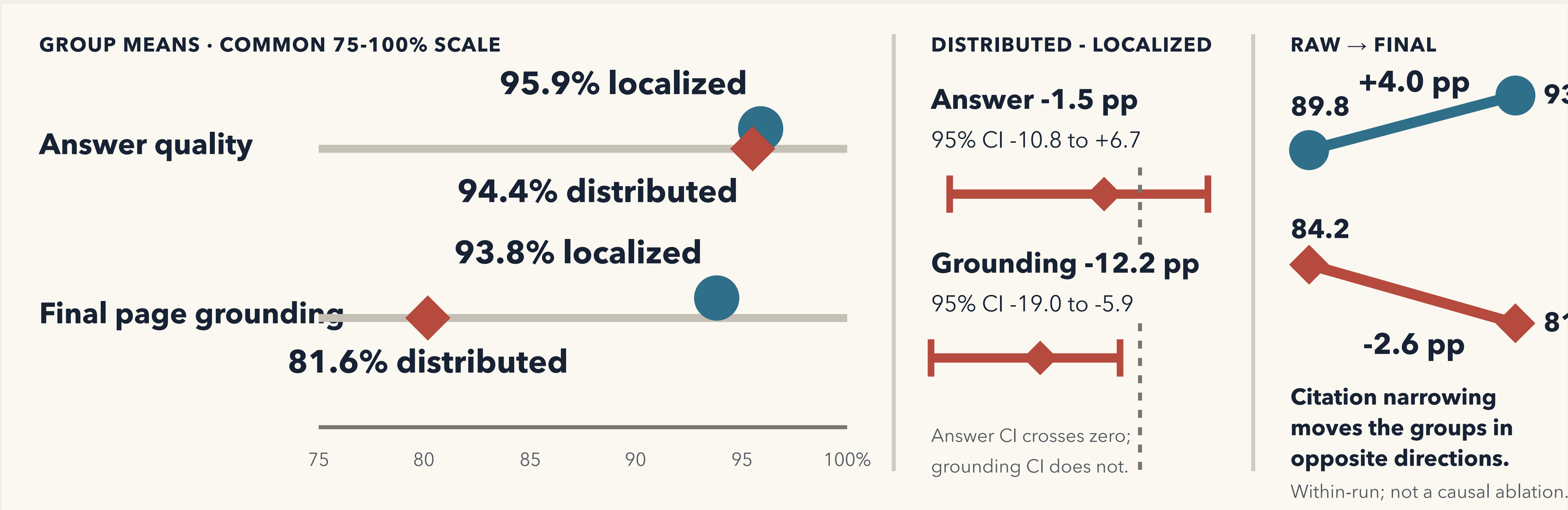
2 METHODOLOGY

30 public DIFC legal PDFs · 590 pages



Every chunk retains document and page identity. 59 vs 36 · 20,000 stratified bootstraps · seed 20260903

3 RESULTS ANSWER QUALITY STAYS CLOSE · GROUNDING SEPARATES



PERFECT ANSWER, GROUNDING < 0.8

● **4 / 55 · 7.3%** ◆ **16 / 34 · 47.1%**

DECISION

Preserve evidence when support is spread.

Route on observable evidence spread or claim-level checks: narrow localized support; aggregate distributed support conservatively. Association consistent with H1; not causal.

LIMITATIONS

- Single run; retrospective hypothesis; observational association.
- Page/document confounding: 35/36 multi-page are multi-document.
- Heterogeneous scoring; uneven answer types; limited external validity.

REPRODUCIBILITY

github.com/SteinVR/Agentic-RAG-Challenge
Run: submission_e2e_20260424_solver_narrowing_no_guard
Lewis 2020 · Cormack 2009 · Gao 2023 · Zhang 2025
Complete references + integrity declaration: appendix